



TLD Group Engineering Guidelines

Rev8

May 25th 2012

The TLD Group is a decentralized organisation where each Industrial entity is fully responsible for the development of the products it has been given the responsibility to engineer and to manufacture. The key local people accountable for the product development are the Engineering Manager and the Chief Operation Officer of the industrial BU.

However, some coordination and controls are necessary at the TLD group level for the following reasons:

- The customer sees and must see TLD Group as a unique and consistent entity
- Similar products can be manufactured in several TLD industrial sites
- Part supports and Cost optimization require standardization above the factory level
- Introduction of any new technology must be done after we insure that it is reliable and the TLD service organization and customers could handle it in the fields

The TLD Group Technical Director (GTD) is in charge of the coordination of all engineering activities of the TLD group.

The TLD Group Technical Director works directly with the manufacturing COO's and with the EM's.

In parallel of validating and controlling some engineering development and proposals engineering group, the GTD role is also to promote innovation, standardization and benchmarking between different TLD factories and insure the development of common processes.

Engineering activity ranking: the different activities of the engineering departments are ranked through the IF (Importance Factor) of the MEAP/EAP system as follow

IF = 1 : Product drawing correction, minor adaptation for better produce-ability or change with insignificant cost impact, and with no direct impact on unit operation and performance

IF = 10: Product on going improvement and customization of a product (Ex: new handrail for a NBL, design of customer specific options ...)

- a. As long as it does not involve new technologies or it does not affect the TLD Standardization or RANGER products, the engineering development is managed by the EM under the supervision of the COO.
- b. If it affects the TLD standardization or the RANGER products, the TLD Group technical director and the other EM's should be consulted before engaging the improvement and the final decision should be approved by all involved parties.
- c. If it applies to the implementation of new components (non critical) which require tests and validations

IF = 100: Significant improvement/evolution/technologic change of a TLD Product line (Ex: Engine evolution or critical component change, new cab evolution, control system change)

Any significant improvement/evolution at TLD needs to be pre-approved by the GTD and then the TLD Group COO.

Once the start of phase 0 of the project has been approved, the project will have to continue to be managed and controlled according to the PACE mode and with the relevant approvals at each phase (table here below).

IF = 1000: New Product Development or complete upgrade of a TLD Product line (Ex: LSP development, redesign of the TF-7-GR)

At TLD, any new product development launch (start of PACE Phase 1) needs to be pre-approved by TLD Group CEO & COO. at this stage, a complete TLD New Product development package will to be filled out (control by the GTD before submission) and will have to contain the information requested to exit Phase 0 (see below chart).

Once the phase 0 of the project has been approved, the project will have to continue to be managed and controlled according to the PACE mode and with the relevant approvals at each phase (table here below).

It is mandatory to open a MEAP for projects type 10, 100 & 1000 when entering Phase 0.

PACE (Product And Cycle-time Excellence) process:

PACE Phase transition	PACE Phase	Phase Description	Mandatory formal approvals for phase transition			
			Dev type #1	Dev type #10	Dev type #100	Dev type #1000
Initiator			Engineer	Engineer, EM, PSM (when Customer specific), COO, PIP moderator	Any from Local CEO, Local COO, PSM, EM, PIP moderator	Any from Group Execom strategic decision, Local COO, PSM, EM, PIP moderator
Validation / Launch of Phase 0			EM or PL*	EM & COO	GTD	GTD
	PHASE 0	Pre-project				
Phase 0 ->1			N/A	N/A	EM, Local COO, local CEO, GTD, Group COO	GTD & TLD Group COO&CFO, Local CEO, COO&CFO or controller
	PHASE 1	Project				
Phase 1->2			N/A	EM	EM, local COO, local CEO, GTD	EM, Local CEO, COO, CFO or controller GTD and Group COO&CFO
	PHASE 2	Engineering				
Phase 2->3			EM or/and PL**	EM	EM, local COO, local CEO, GTD	EM, local CEO, COO& CFO or controller, GTD
	PHASE 3	Prototype				
Phase 3->4			N/A	EM & local COO	EM, local COO, local CEO	EM, local CEO, COO& CFO or controller, GTD and Group COO
	PHASE 4	Industrialisation				
End Project			N/A	N/A	Local COO & local CEO	Local CEO&COO, GTD

*Project leader (PL) can accept modification request, or drawing correction. PL needs to gauge the complexity and the impact of the modification and involve EM when needed.

** PL needs to involve EM for significant modifications

All validations phase highlighted in Yellow should be done through the EAP module and according to the EAP or MEAP rules (see EAP & MEAP help pages).

Phase validation may require a formal meeting. Some project may require some intermediate reviews “critical design review”.

The minimum typical criteria to exit one phase are given in the below chart.

Minimum Criteria to fulfil to Exit each phase				
Phase 0>1	Phase 1>2	Phase 2>3	Phase 3>4	Exit Phase 4
Project Goal and Specification Sheet* DTC (Design To Cost) documentation	Updated Project Schedule General layout and finalized concept drawings Preliminary schematics and flow diagrams Refine spec and performance requirement Control System Design Specification Updated DTC List of deviation to the original target if any List of applicable standard	Updated Project Schedule Drawings set: Structure, weldments, assembly drawing Finalized flow, electric, and hydraulic drawings List of deviation to the validated spec and performances BOM structure Updated DTC Performances prediction Test plan	Updated Project Schedule Major components validations: Engine approval Test reports List of deviation to the validated spec and performances Internal review report (shall include sales and service peoples) Updated DTC List of modification	BOM and Packages review Data base (Doc convergence) Price list Final List of deviation to the validated spec and performances Standard Compliance review Product Configuration and option Sales documentation Unit documentation (Including CH4 with local SPH) Review of Product BP status (excel spreadsheet)
<i>Product BP (excel spreadsheet)(For 1000 -type project only)*</i>	<i>Finalized Product BP (excel spreadsheet)</i>	<i>Review of Product BP deviations (excel spreadsheet)</i>	<i>Review of Product BP deviations (excel spreadsheet)</i>	

* see template

Project Goal and Specification sheet template:

Project Name
Goals and Specification sheet
00/00/00 Rev X

- **Project Goals:**

- **Market:**

- **Main concepts:**

- **Specifications:**

Description:

Control system:

Option(s):

Performance:

Duty cycle:

Applicable Norms and Standards:

Main components:

- **Identified risks:**

- **Market Price:**

- **Quantity** : Estimated Recurrent market

- **Cost Goals:**

- **Development cost:**

Recurring costs:

Non recurring costs:

Rate used for above calculation:

- **Main phase of the development:**

- **Time to market:**

- **Diagrams**

Overall architecture & dimensions (preliminary) :

Visas : TLD Group COO, TLD GTD

Product BP Template (XL Spreadsheet available from Finance):

Company	
Project Name	
Date of Budget	1-Jan-10
Currency	USD
Expected Completion Date Engineering	30/06/2010
Expected Completion Date Prototype	30/06/2010

Maximum Capitalization	
Engineering Costs	-
Prototype Costs	-
Total Capitalization	-

Rate (€ / currency)	
EUR	1
USD	1,45
CAD	1,4336

Engineering Budget	
Budgeted Number of Hours	
Engineering Labour Rate	
Outsourced Engineering expenses	
Material & other costs	
Total Engineering expenses	-

Threshold	58 000
Capitalization	NO

Prototype Budget	
Budgeted Labour Hours	
Burdened Labour Rate	
Total Burdened Labour	-
Material & Services (including FIR)	
Total Prototype Burdened Costs	-

Unit Target Costs	
Labour Hours	
Burdened Labour Rate	-
Total Burdened Labour	-
Material & Services (including FIR)	
Total Target Burdened Costs	-

PROJECT NAME - PRODUCT BUSINESS PLAN

	Net Transfer Price	2010		2011		2012		2013		2014		2016		TOTAL	
		Qty	Revenues	Qty	Revenues	Qty	Revenues	Qty	Revenues	Qty	Revenues	Qty	Revenues	Qty	Revenues
TLD America	104 862		-		-		-		-		-		-	-	-
TLD Asia	104 862		-		-		-		-		-		-	-	-
TLD Europe	104 862		-		-		-		-		-		-	-	-
Total		0	0	0	0	0	0	0	0	0	0	0	0	0	0